

Mohana Kushal Simhadri

Data Scientist | M.Sc. Data Science Student

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ABOUT ME

Data Science master's student at FAU, who learns by building — six end-to-end projects spanning computer vision, deep learning, regression, and deployed data pipelines. Comfortable owning a project from raw data to a working, interpretable, deployed system, including diagnosing and fixing the bugs along the way. Combines analytical problem-solving with a creative approach to shipping data-driven solutions, not just notebooks.

EDUCATION

M.Sc. Data Science

FAU · Oct 2024–Current

Artificial Intelligence, Pattern Recognition and Analysis, Machine Learning

B.Tech Computer Science

VIT Chennai · Aug 2020–Jul 2024

Data Structures & Algorithms, Database Management Systems, Data Visualization, Statistics for Engineers

LANGUAGES

Hindi (native)

English (C1)

German (A2, learning)

SKILLS

Python, SQL, JavaScript

Data Preprocessing, Feature Engineering, Data Pipelines, Model Deployment

Machine Learning, Deep Learning, Predictive Modeling, Computer Vision, CNN, Ensemble Methods (CatBoost, XGBoost, Random Forest), Reinforcement Learning

Scikit-learn, TensorFlow, PyTorch, Keras, Pandas, NumPy, Matplotlib

Git, GitHub Actions (CI/CD), Streamlit

WORK EXPERIENCE

Applied Data Scientist — SmartInternz, India

May 2023 – Jul 2023

- Built predictive models on real-world datasets using Python and R to support business decision-making.
- Handled data preparation and feature engineering, then evaluated model performance and communicated results through data visualization.

PROJECTS

Movement Analysis: SAM 3D Body vs. Marker-Based Mocap (2026)

- Led a university seminar research project testing whether a two-camera markerless setup (Meta's SAM 3D Body) can substitute for a 10-camera marker-based Vicon mocap rig, across 10 trials.
- Built an 8-phase reproducible data pipeline — parsing, per-view pose estimation, Umeyama/Procrustes-based alignment, accuracy metrics, GPA-based two-view fusion, and a literature-grounded synthesis stage — with a full diagnostic and regression-audit suite.
- Root-caused a systematic geometric distortion in the model's hip keypoints, cutting hip/knee flexion-angle RMSE by 66–75% and flipping their correlation with ground truth from strongly negative to strongly positive.

Customized CNNs for Precise Lung Cancer Classification (May 2024)

- Designed and benchmarked three CNN architectures against each other on 25,000 histopathology images across 5 classes: a baseline CNN, a Custom ResNet with attention and ImageNet transfer learning, and a Custom EfficientNet.
- The EfficientNet variant came out on top at 97.2% accuracy (97.1% precision, 97.0% recall, 97.0% F1), ahead of the ResNet (96.8%) and the baseline (94.2%).
- Added interpretability tooling on top — training/validation curves, confusion matrices, Grad-CAM heatmaps — so the model's decisions could actually be inspected, not just trusted.

Novel Crop Yield Prediction Model (Jan 2024)

- Predicting crop yield meant pulling together several disconnected datasets first — yield, rainfall, temperature, and pesticide usage — then enriching them with fertilizer-related features before any modeling could start.
- Evaluated five regression models (Linear Regression, Decision Tree, Random Forest, Gradient Boosting, SVR) using R^2 , MAE, and RMSE. Random Forest Regressor came out ahead.
- Turned the trained model into an interactive Streamlit web app for real-time predictions.

Flight Price Prediction Website (Jul 2023)

- Built the full pipeline for 10,000+ flight records myself: data preparation, temporal feature engineering, and categorical encoding (airline, route, class, stops) via one-hot/label encoding.
- Benchmarked 6 regressors (Random Forest, Gradient Boosting, Decision Tree, KNN, SVR, Logistic Regression) on one 80/20 split; Random Forest led on accuracy at 0.833 R^2 .
- Deployed Gradient Boosting rather than the more accurate forest, and documented the trade-off: the free hosting tier has no backend, so the model is serialised to JavaScript and evaluated in the browser — 0.784 R^2 in 0.03 MB against the forest's 16.2 MB, verified identical to scikit-learn to 8.2e-9.

CERTIFICATIONS

AWS Certified Cloud Practitioner — AWS (Jul 2023)

SmartBridge Externship Certificate – Applied Data Science Powered by Google (Jul 2023)

Fundamentals of Data Analytics — LEAPS by Analyttica (May 2023)

Fundamentals of Data Analytics — NASSCOM FutureSkills (May 2023)

Python Certification — Kaggle (Dec 2022)